MIL-R-46560A (MU)
15 April 1965
SUPERSEDING
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20 July 1961

MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY, M3A1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M3A1, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-413 - Visual Inspection Guide for Rubber O-Rings
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Munitions Command

F8593702 - Remover, Aircraft Canopy, M3A1, Metal Parts Assembly
F8797385 - Container, Ammunition, Fiber, M321 for Remover, Aircraft Canopy, M3A1
F8797386 - Box, Packing, Ammunition for Remover, Aircraft Canopy, M3A1
IEL-8593702 - Index to Inspection Equipment List for Remover, Aircraft Canopy, M3A1, Metal Parts Assembly

FSC 1375

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties.- Physical properties required by the drawings shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure.- The inside and outside tube shall withstand a constant internal pressure of 2700 pounds per square inch (psi), minimum, for a period of 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (See 6.1).

3.6 Shear pin strength.- After application of protective finish, the pin shall shear when a load of 400 $\pm$ 50 pounds is applied at the steady rate of 0.25 inches per minute.

3.7 Structural strength.- The metal parts assembly shall be capable of withstanding each test load shown in Table I for a period of 15 seconds, minimum. The maximum allowable change between mounting points, resulting from each load application, shall not exceed the values specified in Table I. A preload of 500 pounds shall be utilized for each test load application.

Table I

Test Loads (Including 500 pound preload) (min)	Change between mounting points (max)
a. 2000 pounds in tension	.005 inches
b. 2000 pounds in tension	.001 inches
c. 8000 pounds in compression	.005 inches
d. 8000 pounds in compression	.001 inches

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3.8 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawings and the following:

3.8.1 Torque.- Torque requirements shall be as specified in Table II. No loosening or deformation shall occur upon application of the breakaway torque.

Table II

Component	A Run-On (Inch Pounds)	B Seating or Bottoming (Including A) (Inch Pounds)	C Breakaway (Inch Pounds)
Tube bearing and outside tube	75 Min	400 + 50	300 Min
Cap and outside tube	75 Min	500 + 50	400 Min
Block and inside tube	75 Min	---	---
Ring stop and outside tube	75 Min	---	---

3.8.1.1 Bottoming torque.- The bottoming torque between block and inside tube and between ring stop and outside tube shall not exceed the running torque by more than 50 inch pounds.

3.9 Protective finish.- Protective finish shall be in accordance with the requirements of the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.10 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects.- The components shall be free from cracks, splits, cold shuts, inclusions, porosity or similar defects.

3.11.2 Burr.- The component shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 Foreign matter.- The component or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Initial production sample.- At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of parts and 50 shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- The lots of metal parts assemblies shall consist of 614 removers, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

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4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor group the criteria in Table IV shall be used. All nonconforming material shall be rejected. Inspection for O-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table III

Lot Size	Sample Size	Accept Numbers			
		Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

Table IV

Total Allowable Defects Per Sample Size

Sample Size for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over - 43	6	8	11	14
Sample Size for Minor Defects	**110**	**150**	**225**	**300**
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over - 48	8	11	17	20

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4.3.2.1 Latch, Dwg. C8593644, covering a detail of Dwg. F8593702.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Diameter of firing pin relief	Gage
102.	Length to inside surface at small end	Gage
103.	Length of cutout	Gage
104.	Width at small end	Gage
105.	Width at cutout	Gage
106.	Thickness at datum locations	Gage
107.	Location of firing pin relief	Gage
108.	Angle at large end	Gage
109.	Metal defective	Visual
110.	Protective finish inadequate or defective	Visual
111.	Surface finish of base, chamfer and face, improper	Visual
112.	Burr or sharp corners	Visual
113.	Foreign matter	Visual
MINOR		
201.	Total length, max.	Gage
202.	Width at large end	Gage
203.	Thickness, Max.	Gage
204.	Thickness at firing pin relief, max.	Gage

4.3.2.2 Pin, Shear, Dwg. B8593654, covering a detail of Dwg. F8593702.

CRITICAL		
None defined		
MAJOR		
101.	Diameter, max.	Gage
102.	Total length, min.	Gage
103.	Metal defective	Visual
104.	Protective finish, inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Total length, max.	Gage

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4.3.2.3 Pin, Firing, Dwg. C8593657, covering a detail of Dwg. F8593702.

	Defect	Method of Inspection
CRITICAL		
1.	Diameter of striker, max.	Gage
MAJOR		
101.	Concentricity of striker with striker shoulder	Gage
102.	Concentricity of flange with striker shoulder	Gage
103.	Concentricity of groove with flange	Gage
104.	Symmetry of forward datum location with striker shoulder	Gage
105.	Symmetry of width of body with striker shoulder	Gage
106.	Symmetry of thickness of body with striker shoulder	Gage
107.	True position of pin hole with striker shoulder and thickness of body	Gage
108.	Diameter of striker shoulder, min.	Gage
109.	Diameter of flange, min.	Gage
110.	Diameter of groove	Gage
111.	Depth of hole in striker shoulder, max.	Gage
112.	Length of striker, min.	Gage
113.	Width of body	Gage
114.	Width of groove	Measure
115.	Relief groove on striker shoulder missing	Visual
116.	Metal defective	Visual
117.	Protective finish inadequate or defective	Visual
118.	Surface finish of body, flange and striker shoulder, improper	Visual
119.	Burr or sharp corner except as permitted	Visual
120.	Foreign matter	Visual
MINOR		
201.	Diameter of striker shoulder, max.	Gage
202.	Diameter of flange, max.	Gage
203.	Diameter of hole in striker shoulder	Gage
204.	Depth of hole in striker shoulder, min.	Gage
205.	Width at forward datum location	Gage
206.	Width at rear datum location	Gage
207.	Location of hole	Gage
208.	Surface variation on rear face of striker shoulder	Gage

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4.3.2.4 Seal, Fixed, Dwg. B8593660, covering a detail of Dwg. F8593702.

Defect		Method of Inspection
CRITICAL		
None defined		
MAJOR		
101.	Diameter of wire, max. (after plating)	Gage
102.	Length of tang	Gage
103.	Location variance of ends, max.	Visual
104.	Number of coils, incorrect	Visual
105.	Radius of tang bend, max.	Visual
106.	Metal defective	Visual
107.	Protective finish inadequate or defective	Visual
108.	Burr or sharp edge	Visual
109.	Foreign matter	Visual
MINOR		
201.	Diameter, outside	Gage
202.	Direction of turn, incorrect	Visual

4.3.2.5 Cap, Dwg. D8593703, covering a detail of Dwg. F8593702.

CRITICAL		
None defined		
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Concentricity of body with pitch diameter of thread	Gage
104.	Symmetry of width of lug with pitch diameter of thread	Gage
105.	Symmetry of thickness of lug with pitch diameter of thread	Gage
106.	True position of hole with pitch diameter of thread including perpendicularity with lug	Gage
107.	Diameter of body, min.	Gage
108.	Diameter of thread undercut, max.	Gage
109.	Diameter of hole	Gage
110.	Width of lug	Gage
111.	Thickness of lug	Gage
112.	Thickness of base, min.	Gage
113.	Thread damaged or not full	Visual
114.	Metal defective	Visual
115.	Protective finish inadequate or defective	Visual
116.	Burr or sharp corner	Visual
117.	Foreign matter	Visual

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Defect		Method of Inspection
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Diameter of body, max.	Gage
203.	Location of hole	Gage
204.	Depth of cavity	Measure
205.	Total length	Measure
206.	Width of thread undercut, max.	Measure
207.	Profile of mounting lug	Measure
208.	Marking missing, incorrect or illegible	Visual
209.	Radius or chamfer missing or incorrect	Visual
210.	Surface finish improper	Visual

4.3.2.6 Tube, Bearing, Dwgs. D8593706, C8593707, covering a detail of Dwg. F8593702.

CRITICAL		
None defined		
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of small bore with large bore	Gage
104.	Concentricity of body with pitch diameter of thread	Gage
105.	Concentricity of large bore with pitch diameter of thread	Gage
106.	Concentricity of counterbore with pitch diameter of thread	Gage
107.	Diameter of small bore	Gage
108.	Diameter of counterbore, max.	Gage
109.	Depth of spiral groove	Gage
110.	Thickness of wall under pellet	Gage
111.	Pellet missing, loose or improperly inserted	Visual/Manual
112.	Metal defective	Visual
113.	Protective finish inadequate or defective	Visual
114.	Surface finish on small bore, improper	Visual
115.	Burr or sharp corner	Visual
116.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of large bore	Gage
203.	Diameter of counterbore, min.	Gage
204.	Depth of spiral groove relief	Gage
205.	Depth to bottom of large bore	Gage

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Defect	Method of Inspection
MINOR (Cont'd)	
206. Diameter of small hole	Measure
207. Depth of counterbore	Measure
208. Length of thread	Measure
209. Location of spiral groove	Measure
210. Symmetry of tang hole with spiral groove	Visual
211. Length of relief groove	Visual
212. Width of relief groove	Visual
213. Location of relief groove	Visual
214. Location of slot	Visual
215. Direction of spiral groove turn, incorrect	Visual
216. Number of spiral groove turns, incorrect	Visual

4.3.2.7 Ring, Stop, Dwg. C8593708, covering a detail of Dwg. F8593702.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of bore with pitch diameter of thread	Gage
104. Concentricity of counterbore with bore	Gage
105. Symmetry of width across flats with flange	Gage
106. Diameter of thread undercut, min.	Gage
107. Diameter of bore	Gage
108. Depth of pellet hole	Gage
109. Length to counterbore	Gage
110. Location to datum diameter	Gage
111. Length to flange	Gage
112. Angle at datum diameter	Gage
113. Thickness of wall under pellet	Gage
114. Pellet missing, loose or improperly inserted	Visual/Manual
115. Thread damaged or not full	Visual
116. Metal defective	Visual
117. Protective finish inadequate or defective	Visual
118. Surface finish or angle of bore or bore	Visual
119. Burr or sharp corner	Visual
120. Foreign matter	Visual
MINOR	
201. Total Length	Gage
202. Pitch diameter of thread, max.	Gage
203. Diameter of bore, min.	Gage
204. Diameter of counterbore	Gage
205. Diameter of flange	Gage

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Defect		Method of Inspection

4.3.2.8 Tube, Outside, Dwgs. D8593710, C8593711, covering a detail of Dwg. F8593702.

CRITICAL
None defined

	MAJOR	
101.	Pitch diameter of external thread, min.	Gage
102.	Major diameter of external thread, min.	Gage
103.	Pitch diameter of internal thread, max.	Gage
104.	Minor diameter of internal thread, max.	Gage
105.	Concentricity of pitch diameter of external thread with inside diameter	Gage
106.	Concentricity of pitch diameter of internal thread with inside diameter	Gage
107.	Concentricity of large outside diameter with pitch diameter of internal thread	Gage
108.	Diameter of body, min.	Gage
109.	Diameter of groove, min.	Gage
110.	Diameter of undercut, max.	Gage
111.	Diameter of inside, max.	Gage
112.	Length of effective external thread, min.	Measure
113.	Wall thickness, min.	Gage
114.	Straightness	Gage
115.	Thickness of wall under pellet	Gage
116.	Pellet missing, loose or improperly inserted	Visual/Manual
117.	Metal defective	Visual
118.	Protective finish inadequate or defective	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual

	MINOR	
201.	Pitch diameter of external thread, max.	Gage
202.	Pitch diameter of internal thread, min.	Gage
203.	Length of thread, max.	Gage
204.	Diameter, inside, min.	Gage
205.	Total length, max.	Gage
206.	Location of groove	Gage
207.	Radius or chamfer missing or incorrect	Visual
208.	Marking missing, incorrect or illegible	Visual
209.	Surface finish improper	Visual

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Defect		Method of Inspection

4.3.2.9 Tube, Inside, Dwgs. D8593712, C8593713, covering a detail of Dwg. F8593712.

CRITICAL
None defined

MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of pitch diameter of thread with counterbore	Gage
104.	Concentricity of counterbore with small inside diameter	Gage
105.	Diameter of body	Gage
106.	Diameter of counterbore, max.	Gage
107.	Diameter, inside, small end, max.	Gage
108.	Diameter, inside, large end, max.	Gage
109.	Depth of counterbore, max.	Gage
110.	Length of effective thread, min.	Gage
111.	Thickness of wall, min. small end	Gage
112.	Thickness of wall, min. large end	Gage
113.	Thickness of wall under pellet	Gage
114.	Straightness	Gage
115.	Pellet missing, loose or improperly assembled	Visual
116.	Metal defective	Visual
117.	Protective finish inadequate or defective	Visual
118.	Surface finish on body, improper	Visual
119.	Burr	Visual
120.	Foreign matter	Visual

MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of counterbore, min.	Gage
203.	Diameter inside, small end, min.	Gage
204.	Diameter inside, large end, min.	Gage
205.	Depth of counterbore, min.	Gage
206.	Depth of inside diameter, large end	Gage
207.	Total length	Gage
208.	Length of thread, max.	Measure
209.	Radius or chamfer missing or incorrect	Visual
210.	Surface finish, improper, except as otherwise classified	Visual

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Defect	Method of Inspection

4.3.2.10 Block, Dwg. D8593714, covering a detail of Dwg. F8593702.

CRITICAL
None defined

MAJOR		
101.	Pitch diameter of pipe thread	Gage
102.	Minor diameter of pipe thread, max.	Gage
103.	Pitch diameter of thread, max.	Gage
104.	Minor diameter of thread, max.	Gage
105.	Concentricity of counterbore with small cavity and pitch diameter of thread	Gage
106.	Concentricity of large body with pitch diamater of thread	Gage
107.	Concentricity of flange with body	Gage
108.	Symmetry of slot with body	Gage
109.	Symmetry of thickness of rectangular section with body	Gage
110.	True position of screw hole with counterbore and base of counterbore	Gage
111.	True position of mounting holes with large body including perpendicularity with mounting lug	Gage
112.	True position of pin hole with small cavity	Gage
113.	Diameter of mounting hole	Gage
114.	Diameter of small cavity	Gage
115.	Diameter of body at small cavity area	Gage
116.	Diameter of undercut, max.	Gage
117.	Diameter of counterbore, max.	Gage
118.	Depth of chamfer at small cavity	Gage
119.	Depth of threaded cavity including undercut, min.	Gage
120.	Depth to bottom of cavity	Gage
121.	Depth to bottom of drilled hole, max.	Gage
122.	Thickness of rectangular section	Gage
123.	Length to flange	Gage
124.	Location of slot to base of counterbore	Gage
125.	Location of pipe thread hole	Measure
126.	Thread damaged or not full	Visual
127.	Metal defective	Visual
128.	Protective finish inadequate or defective	Visual
129.	Surface finish of small cavity, body and slot, improper	Visual
130.	Burr	Visual
131.	Foreign matter	Visual

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	Defect	Method of Inspection
MINOR		
201.	Pitch diameter of thread min.	Gage
202.	Pitch diameter of pipe thread	Gage
203.	Diameter of flange max.	Gage
204.	Diameter of counterbore, min.	Gage
205.	Diameter of body	Gage
206.	Diameter of pin hole	Gage
207.	Diameter of screw hole	Gage
208.	Diameter of mounting hole	Gage
209.	Length of body including flange	Gage
210.	Length of slot in body	Gage
211.	Width of slot in body	Gage
212.	Location of screw hole and angular location with slot	Gage
213.	Location of mounting hole	Gage
214.	Location of pin hole and angular location with slot	Gage
215.	Depth to bottom of small cavity	Gage
216.	Diameter of small cavity	Measure
217.	Total length	Gage
218.	Profile of end of rectangular section	Measure

4.3.2.11 Plug, Dwg. C8593715, covering a detail of Dwg. F8593702.

CRITICAL		
1.	Diameter of boss hole, min.	Gage
MAJOR		
101.	Concentricity of boss hole with cavity	Gage
102.	Concentricity of flange with cavity	Gage
103.	Concentricity of body with flange	Gage
104.	Symmetry of base slot with cavity	Gage
105.	True position of pin hole with base slot and cavity	Gage
106.	True position of pitch diameter of thread with flange including perpendicularity with base slot	Gage
107.	Diameter of cavity	Gage
108.	Diameter of flange, min.	Gage
109.	Depth of threaded hole, max.	Gage
110.	Depth of cavity, min.	Gage
111.	Depth of slot	Gage
112.	Total length	Gage
113.	Height of boss	Gage
114.	Chamfer in boss hole, improper	Gage
115.	Metal defective	Visual
116.	Protective finish inadequate or defective	Visual
117.	Surface finish of cavity and slot, improper	Visual
118.	Burr or sharp corner	Visual
119.	Foreign matter	Visual

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Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread	Gage
202. Minor diameter of thread, max.	Gage
203. Diameter of flange, max.	Gage
204. Diameter of body	Gage
205. Diameter of pin hole, min.	Gage
206. Depth of cavity, max.	Gage
207. Length of effective thread, min. including chamfer	Gage
208. Width of slot	Gage
209. Location of pin hole	Gage
210. Location of threaded hole	Gage
211. Diameter of boss	Measure
212. Length to shoulder	Measure
213. Marking missing, incorrect or illegible	Visual
214. Surface finish improper except as otherwise classified	Visual

4.3.2.12 Metal Parts Assembly, Dwg. F8593702.

CRITICAL	
None defined	
MAJOR	
101. Any part missing, except as otherwise classified	Manual/Visual
102. Improper assembly	Visual
103. Thread damaged or not full	Visual
104. Marking missing, incorrect or illegible	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Shipping plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.3.2.13 Packing in fiber container, Dwg. F8797385.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Cap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

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4.3.2.14 Packing in Wooden Box, Dwg. F8797386.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested to determine compliance with the requirements of the drawings. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 Hydrostatic pressure.- Each inside and outside tube shall be subjected to an internal hydrostatic pressure. Failure of any tube to comply with the requirements of 3.5 shall be cause for rejection of the tube.

4.3.3.3 Shear pin strength.- Fifty pins from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.3.4 Structural strength.- A sample of 15 complete assemblies shall be selected and subjected to the structural strength test. Failure of any assembly to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.3.3.5 Breakaway torque.- A breakaway torque shall be applied to each assembly threaded joint as listed in Table II. Failure of any assembly to comply with the requirements of Column "C" of Table II shall be cause for rejection of the assembly.

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4.3.3.6 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing.

4.3.3.7 Residual magnetism.- Each metal parts assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593702.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the detail drawings.

4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the inside and outside tubes shall be mounted in the test device and subjected to an internal hydrostatic pressure to determine compliance with the requirements of 3.5. The test device used shall internally seal the tube within 1-1/2 inches of each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.- After application of protective finish, the pin shall be subjected to the shear strength test to determine compliance with the requirements of 3.6. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.4.4 Structural strength.- The remover shall be placed in the tensile testing machine and subjected to the specified tension and compression loads for determination of compliance with the requirements of 3.7. The test shall be performed as follows:

a. Apply a preload of 500 pounds so as to move the mounting points away from each other and adjust the dial indicator to zero inches.

b. Apply the specified load in tension.

c. Decrease the load to 500 pounds and record the dial indicator reading.

d. Repeat steps "b" and "c"

e. Apply a preload of 500 pounds so as to move the mounting points toward each other and adjust the dial indicator to zero inches.

f. Apply the specified load in compression.

g. Decrease the load to 500 pounds and record the dial indicator reading.

h. Repeat steps "f" and "g".

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4.4.5 Breakaway torque.- A torque wrench set at the torque requirements specified in 3.8.1 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to the application of the torque.

4.4.6 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.7 Residual magnetism.- The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly 6 inches from and at the same height as the compass, in a north-south horizontal position (block end at south) with the compass set in an east-west heading as shown in Figure I. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length then rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.- Packing and marking shall be in accordance with Drawings F8797385 and F8797386.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

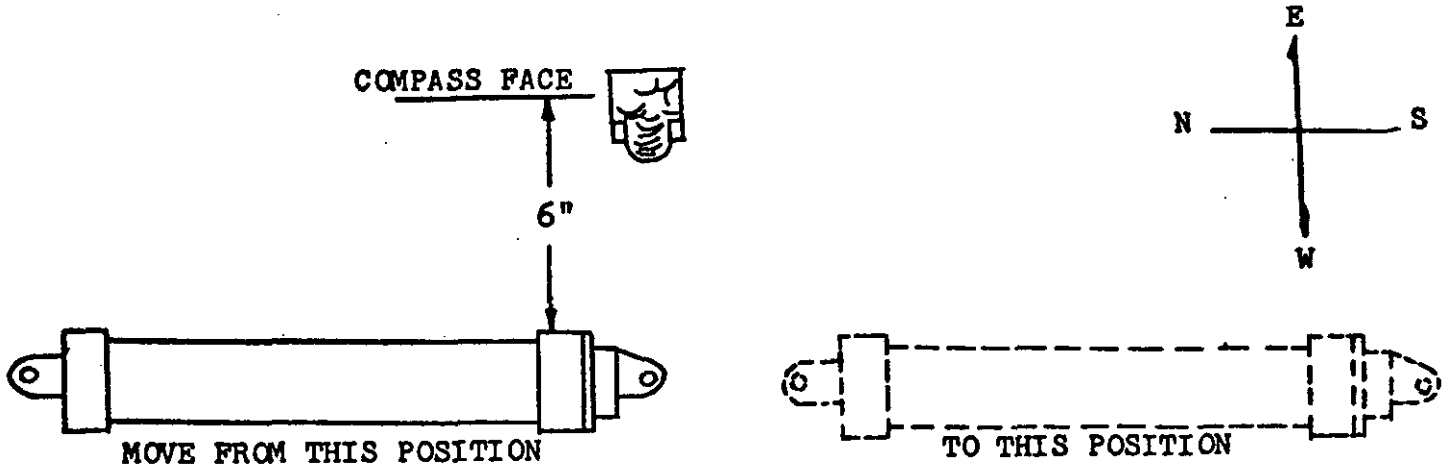
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PILOT'S STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

NOTE:

Remover and compass shall be in a horizontal plane

Figure 1

Layout for determination of residual magnetism in Remover, Aircraft Canopy, M3A1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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